The documentation and process conversion measures necessary to comply with this revision shall be completed by 16 January 2012.

INCH-POUND
MIL-PRF-19500/37E
w/AMENDMENT 1
16 December 2011
SUPERSEDING
MIL-S-19500/37E
16 September 2001

* PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER TYPES 2N333, 2N335, 2N336, 2N333A, 2N335A, 2N336A, 2N333T2, 2N335T2, 2N336T2, 2N333LT2, 2N335LT2, 2N336LT2 2N333AT2, 2N335AT2, 2N336AT2, 2N333ALT2, 2N335ALT2, AND 2N336ALT2, JAN

Inactive for new design after 7 June 1999

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, low-power transistors.

1.2 Physical dimensions. See figure 1 (TO-5 for isolated leads only) and figure 2 (TO-205AA).

1.3 Maximum ratings.

P_T T_A = 25°C		V_{CEO}	V_{CBO}	V_{EBO}		T_{stg}	
2N333, T2, LT2 2N335, T2, LT2 2N336, T2, LT2	2N333A, T2, LT2 2N335A, T2, LT2 2N336A, T2, LT2			2N333, T2, LT2 2N335, T2, LT2 2N336, T2, LT2	2N333A, T2, LT2 2N335A, T2, LT2 2N336A, T2, LT2	2N333, T2, LT2 2N335, T2, LT2 2N336, T2, LT2	2N333A, T2, LT2 2N335A, T2, LT2 2N336A, T2, LT2
mW	mW	V dc	V dc	V dc	V dc	°C	°C
(1) 150	(2) 500	45	45	1	4	-65 to +175	-65 to +175

(1) Derate approximately 1 mW/°C for T_A between +25°C and +175°C.
(2) Derate approximately 3.33 mW/°C for T_A between +25°C and +175°C.

1.4 Primary electrical characteristics.

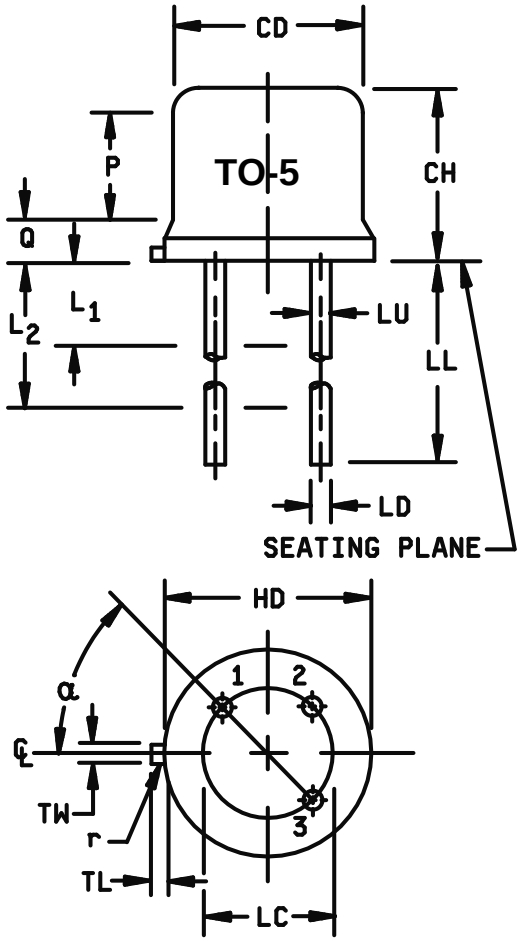
Limits	h_{fe} V_{CB} = 5 V dc, I_E = -1 mA			f_{hfb} V_{CB} = 5 V dc I_E = -1 mA dc	C_{obo} V_{CB} = 5 V dc I_E = 0 mA dc 100kHz ≤ f ≤ 1 MHz
	2N333, T2, LT2 2N333A, T2, LT2	2N335, T2, LT2 2N335A, T2, LT2	2N336, T2, LT2 2N336A, T2, LT2		
				MHz	pF
Min	18	37	76	2.5	
Max	44	90	270		15

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

AMSC N/A FSC 5961

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		6
LD	.016	.021	0.41	0.53	7, 8
LL	1.500	1.750	38.10	44.45	7, 8
LU	.016	.019	0.41	0.48	7, 8
L_1		.050		1.27	7, 8
L_2	.250		6.35		7, 8
Q		.050		1.27	5
TL	.029	.045	0.74	1.14	3, 4
TW	.028	.034	0.71	0.86	3
r		.010		0.25	10
α	45°TP		45°TP		6

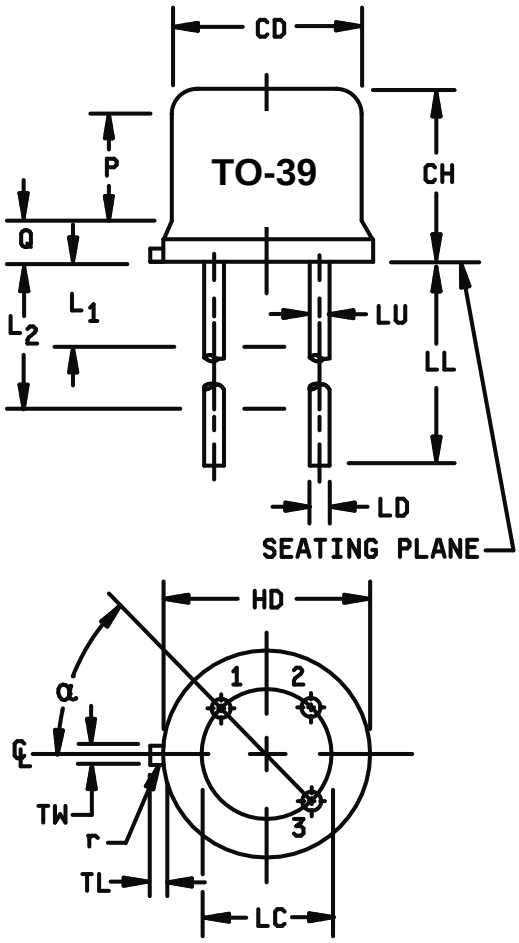


NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All leads electrically isolated from the case.
9. Dimension r (radius) applies to both inside corners of tab.
10. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 1. Physical dimensions of transistors, types 2N333, 2N333A, 2N335, 2N335A, 2N336, and 2N336A (TO-5).

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		6
LD	.016	.021	0.41	0.53	7, 8
LL	.500	.750	12.70	19.05	7, 8, 12
LU	.016	.019	0.41	0.48	7, 8
L_1		.050		1.27	7, 8
L_2	.250		6.35		7, 8
Q		.050		1.27	5
TL	.029	.045	0.74	1.14	3, 4
TW	.028	.034	0.71	0.86	3
r		.010		0.25	10
α	45°TP		45°TP		6



NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with AMSE Y14.5M, diameters are equivalent to Φx symbology.
12. For "L" suffix devices, dimension LL is 1.50 (38.10 mm) minimum, 1.75 (19.05 mm) maximum.

FIGURE 2. Physical dimensions of transistors, types 2N333T2, 2N333AT2, 2N335T2, 2N335AT2, 2N336T2, 2N336AT2 2N333LT2, 2N333ALT2, 2N335LT2, 2N335ALT2, 2N336LT2, and 2N336ALT2 (TO-205AD).

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2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch or https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 and figure 2.

* 3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and tables I, II, and III.

* 3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I.

* 3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

* 4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.3 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.3.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

* 4.3.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIb (JAN) of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with table II herein, as applicable.

* 4.3.3 Group C inspection. Group C inspection shall be conducted in accordance with the test and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with table III herein, as applicable.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Details		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 30 V dc	I_{CEO}		1.0	µA dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO1}		0.5	µA dc
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 50 µA dc	BV_{CBO}	45		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 1 mA dc	BV_{CEO1}	45		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 100 µA dc	BV_{CEO2}	45		V dc
Breakdown voltage, emitter to base	3026	Bias condition D	BV_{EBO}			
2N333, 2N335, 2N336, T2, LT2 2N333A, 2N335A, 2N336A, T2, LT2		I_E = 10 µA dc I_E = 100 µA dc		1.0 4.0		V dc V dc
Collector to emitter voltage (saturated)		I_C = 5 mA dc	$V_{CE(sat)}$			
2N333, 2N335, 2N336, T2, LT2 2N333A, 2N335A, 2N336A, T2, LT2		I_B = 2.2 mA dc I_B = 1.0 mA dc			1.0 1.0	V dc V dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 mA dc	h_{FE1}			
2N333, 2N333A, T2, LT2 2N335 2N335A, T2, LT2 2N336, 2N336A, T2, LT2				6 12 25	60 120 275	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 100 µA dc	h_{FE2}			
2N333, 2N333A, T2, LT2 2N335 2N335A, T2, LT2 2N336, 2N336A, T2, LT2				3 6 15		

See footnote at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Details		Min	Max	
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO2}		25	µA dc
Low-temperature operation:		T_A = -65°C				
Small-signal short-circuit forward-current transfer ratio	3206	V_{CB} = 5 V dc I_E = -1 mA dc	h_{FE3}			
2N333, 2N333A, T2, LT2 2N335 2N335A, T2, LT2 2N336, 2N336A, T2, LT2				8 18 35		
Subgroup 4						
Small-signal open-circuit output admittance	3216	V_{CB} = 5 V dc; I_E = -1 mA dc	h_{ob}	0	1.2	µmho
Small-signal open-circuit reverse-voltage transfer ratio	3211	V_{CB} = 5 V dc; I_E = -1 mA dc	h_{rb}	0	$1x10^{-3}$	
Small-signal open-circuit input impedance	3201	V_{CB} = 5 V dc; I_E = -1 mA dc	h_{ib}			
2N333, 2N333A 2N335, 2N335A 2N336, 2N336A				30	80	ohms
For T2, and LT2 only.				20	80	ohms
Small-signal open-circuit forward-current transfer ratio	3206	V_{CB} = 5 V dc; I_E = -1 mA dc	h_{FE4}			
2N333, 2N333A, T2, LT2 2N335, 2N335A, T2, LT2 2N336, 2N336A, T2, LT2				18 37 76	44 90 270	
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 5 V dc; I_C = 1 mA dc; f = 2.5 MHz	$\lvert h_{fe1} \rvert$			
2N333, T2, LT2 2N335, T2, LT2 2N336, T2, LT2 2N333A, T2, LT2 2N335A, T2, LT2 2N336A, T2, LT2					40 60 120 30 30 30	

See footnote at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Details		Min	Max	
Subgroup 4						
Magnitude of common-emitter small-signal short circuit forward-current transfer ratio	3306	V_{CE} = 5 V dc; I_C = 1 mA dc; f = 10 MHz	$\lvert h_{fe2} \rvert$			
2N333, T2, LT2					13	
2N335, T2, LT2					15	
2N336, T2, LT2					17	
2N333A, T2, LT2					13	
2N335A, T2, LT2					13	
2N336A, T2, LT2					13	
Small-signal short-circuit forward-current transfer ratio, cutoff frequency	3301	V_{CB} = 5 V dc; I_E = -1 mA dc	f_{hfb}	2.5		MHz
Subgroup 5						
Open circuit output capacitance	3236	V_{CB} = 5 V dc; V_{sig} = 0.1 Vac; 100 kHz ≤ f ≤ 1MHz	C_{obo}		15	pF
Noise figure	3246	V_{CB} = 5 V dc; I_E = -1 mA dc; f = 1kHz; R_g = 500 ohms	NF		30	dB

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Group B inspection.

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Details		Min	Max	
Subgroup 1						
Physical dimensions	2066	(See figure 1)				
Subgroup 2						
Solderability	2026					
Temperature cycling (air to air)	1051	Test condition C, except: 10 cycles; exposure time at temperature extremes = 15 minutes (minimum); and in step 3, T_{max} = +175°C.				
Thermal shock (liquid to liquid)	1056	Test condition B				
Moisture resistance	1021					
End points:						
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO}		0.5	µA dc
Small-signal short-circuit forward-current transfer ratio	3206	V_{CB} = 5 V dc; I_E = -1 mA dc	h_{fe}			
2N333, 2N333A, T2, LT2				18	44	
2N335 2N335A, T2, LT2				37	90	
2N336, 2N336A, T2, LT2				76	270	
Subgroup 3						
Shock (2N333, 2N335, 2N336, T2, LT2)	2016	Non-operating; 500 G; 1 ms; 5 blows in each orientation X_1, Y_1, Y_2, and Z_1.				
Shock (2N333A, 2N335A, 2N336A, T2, LT2)	2016	Non-operating; 1500 G; approximately 0.5 ms; 5 blows in each orientation X_1, Y_1, Y_2, and Z_1.				
Vibration, variable frequency	2056	Non-operating				
Constant acceleration	2006	20,000 G in each orientation X_1, Y_1, Y_2, and Z_1				
End points (same as subgroup 2)						

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TABLE II. Group B inspection - Continued.

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Details		Min	Max	
Subgroup 4						
Terminal strength (lead fatigue)	2036	Test condition E				
Hermetic seal	1071	.				
Subgroup 5						
Salt atmosphere (corrosion)	1041					
Subgroup 6						
High temperature life (non-operating) 2N333, 2N335, 2N336, T2, LT2 2N333A, 2N335A, 2N336A, T2, LT2	1032	Time = 340 hours T_{tg} = +175°C T_{tg} = +200°C				
End points:						
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO}		1.0	µA dc
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 50 µA dc	BV_{CBO}	40		V dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 mA dc	Δh_{FE}		±25	Percent change from initial recorded value
Small-signal short-circuit forward current transfer ratio	3206	V_{CB} = 5 V dc; I_E = -1 mA dc	h_{fe}			
2N333, 2N333A, T2, LT2				15		
2N335 2N335A, T2, LT2				30		
2N336, 2N336A, T2, LT2				60		
Subgroup 7						
Steady-state operation life (2N333, 2N335, 2N336, T2, LT2)	1027	T_A = +125°C; V_{CB} = 20 V dc; P = 50 mW; time = 340 hours				
End points: (same as subgroup 6)						
Steady-state operation life (2N333A, 2N335A, 2N336A, T2, LT2)	1027	T_A = +25°C; V_{CB} = 30 V dc; P = 500 mW; time = 340 hours				
End points: (same as subgroup 6)						

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* TABLE III. Group C inspection.

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Details		Min	Max	
Subgroup 1						
Resistance to solvents	1022					
Subgroup 2						
High temperature life (non-operating) 2N333, 2N335, 2N336, T2, LT2 2N333A, 2N335A, 2N336A, T2, LT2	1031	T_{tsg} = +175°C T_{tsg} = +200°C				
End points: (same as subgroup 6 of group B)						
Subgroup 3						
Steady-state operation life 2N333, 2N335, 2N336, T2, LT2	1026	T_A = +125°C; V_{CB} = 20 V dc; P = 50 mW				
End points: (same as subgroup 6 of group B)						
Steady-state operation life 2N333A, 2N335A, 2N336A, T2, LT2	1026	T_A = +25°C; V_{CB} = 30 V dc; P = 500 mW				
End points: (same as subgroup 6 of group B)						

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

* 6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.daps.dla.mil .

* 6.4 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

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